

Data Drift Observation

in a Stock Prediction Model

Project Stock-ML — MLOps Pipeline & Drift Analysis

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The Challenge: Predicting the Unpredictable

Input

Last 30 days of closing prices variation

Output

Predict the variation for the next 5 days

Data

20 tech stocks with 5 years of history

Understand how models degrade over time and study data drift across multiple generations.

Concept Drift and Data Drift

1

Concept Drift

The relationship between X and Y transforms

- New macroeconomic effects
- New regulations

2

Data Drift - Feature Drift

Input variables change

- Market volatility increases
- Stock Volume change

3

Data Drift - Label Drift

The target changes

- New price dynamics
- Prices increase overall (inflation)

Consequences

Gradual performance degradation and risk of erroneous financial decisions.



OUR PROJECT ARCHITECTURE

1 - Data Acquisition



Collection of historical data (5 years) of 20 stocks in the NASDAQ index. Data in CSV format.

2 - Data Preprocessing



Data cleaning and transformation in a format suitable for modeling.

3 - Data Storage



Data is stored in a S3 bucket.

4 - Model training



Creation of a Long-short Term Memory (LSTM) neural network, trained with the preprocessed data.

5 - Model evaluation and comparaison



Evaluation of the model using different metrics and comparing results with other models.

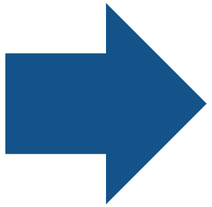
Our MLOps Architecture

CI/CD Pipeline Overview

GitHub Actions, DVC, CML Report, BentoML

Train & Report

- Setup environment
- Run model training via DVC
- Push updated artifacts and metrics to DVC remote
- Generate CML report with:
  - Parameter & metric diffs
  - Plot comparisons (training, predictions)
- Post report as PR comment



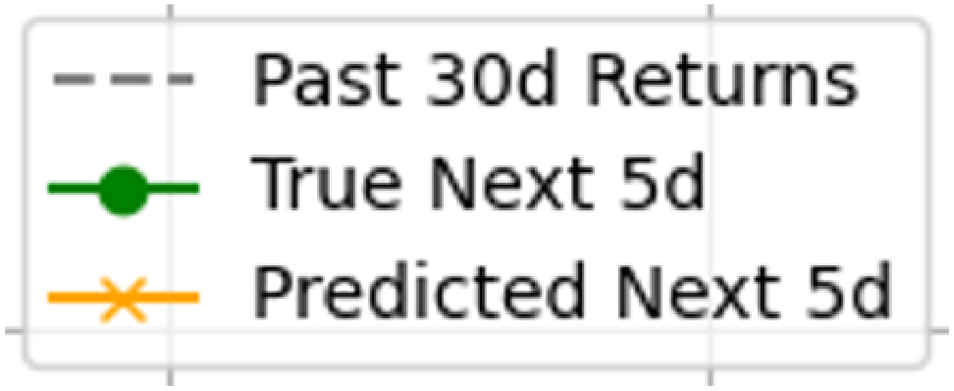
Publish & Deploy

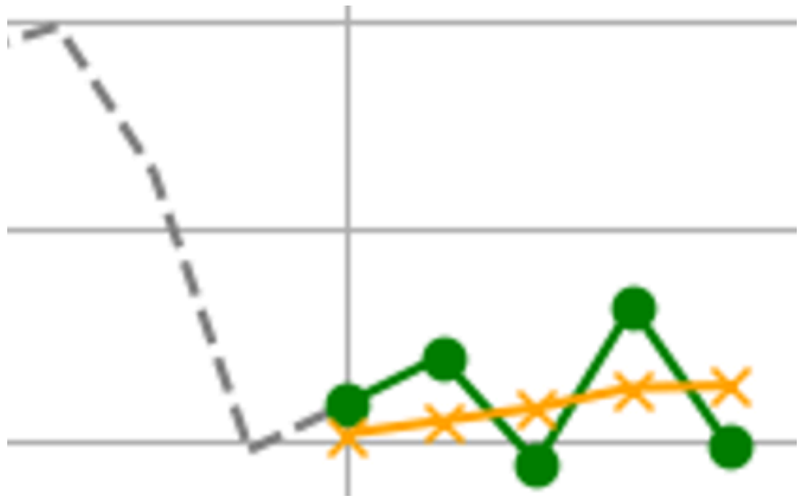
- Build and package BentoML model
- Push Docker image
- Update & apply GKE deployment manifests
- Automatically deploy new model version to Kubernetes

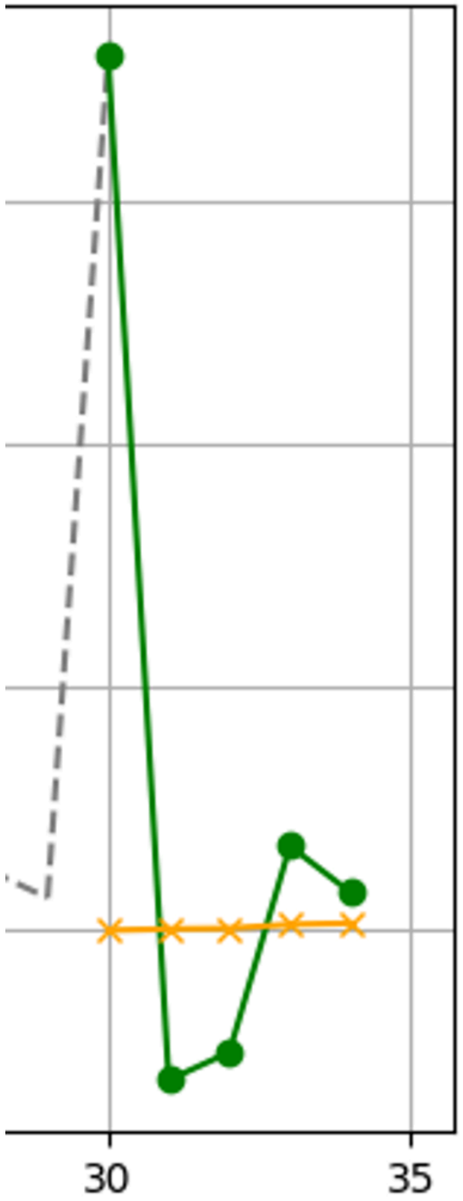
Current Results

- **MAE: 1.87**
- **RMSE: 2.87**
- **R2: 0.004**

The model’s predictions are equivalent to using the mean as the prediction







Actual challenges & next steps

A breakthrought of actual challenges and steps that will be taken in the following weeks

Bad model predictions

Inaccurate predictions (due to volatility of financial markets)

Adaptation of our model target

Changes in our prediction goal (maybe predict trend direction (up/down) or percentage variation).

Data drift evaluation

Monitor our model’s performance over time with new weekly financial data added.



Conclusions & Perspectives

A model that performs well today won't necessarily do so tomorrow. Drift monitoring is the key to sustainable MLOps.

Inevitability

Data drift is inevitable in financial data

Impact

It degrades models without periodic updates

Solution

DVC + versioning allows tracking this impact

Outlook

Automatic monitoring and weekly retraining



Thank you for your attention!